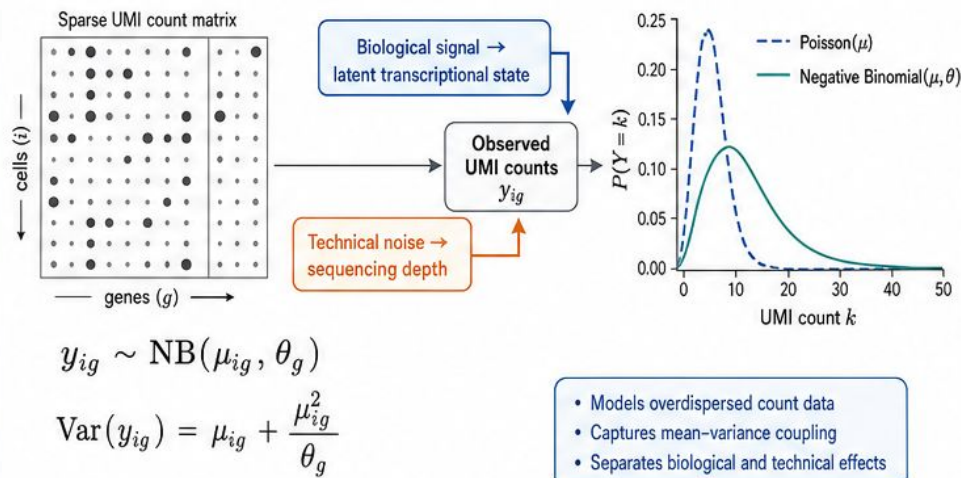
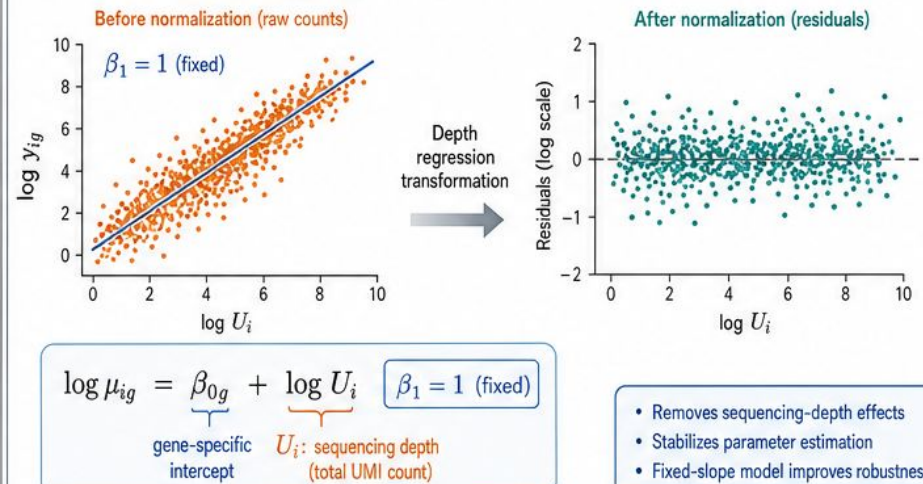


SCTransform in Seurat v5: regularized negative binomial normalization and variance stabilization

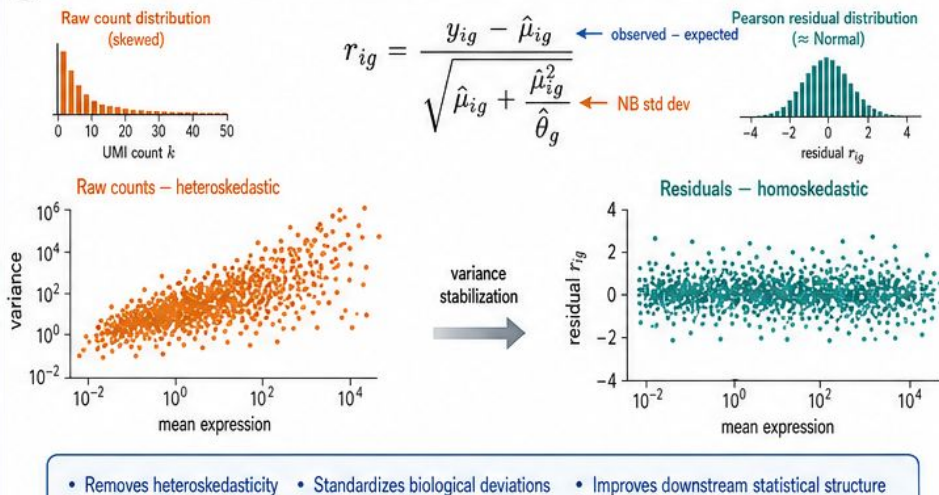
A Negative Binomial Observation Model



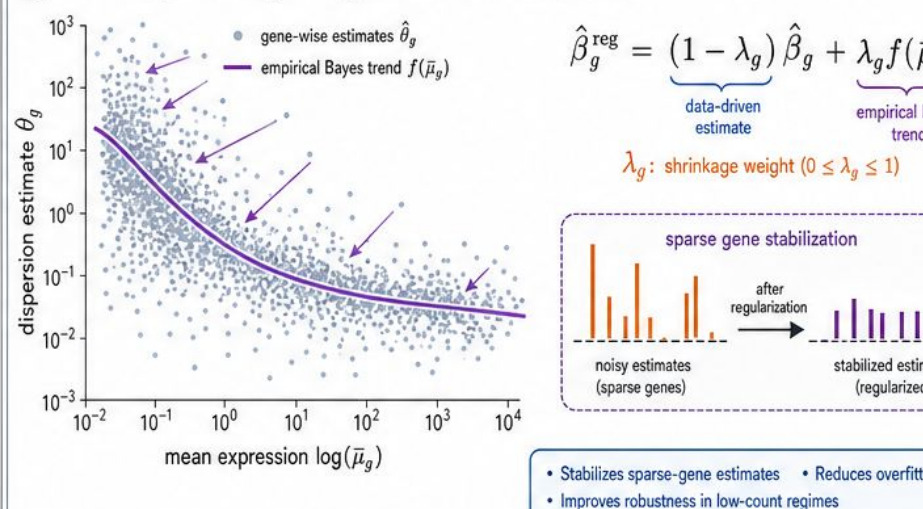
B Sequencing Depth Regression



C Pearson Residual Estimation and Variance Stabilization



D Empirical Bayes Regularization Across Genes



— raw / biological — normalized / residual — regularized / empirical Bayes — model fit